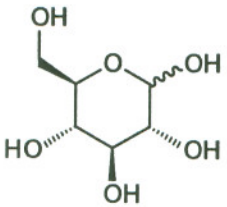
*Analytical Chemistry Cumulative Exam*
February 5, 2010

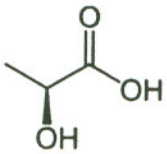
**Instructions:**
Pick **two problems** out of the four listed. For these two problems, answer the questions keeping in mind that grading will be based on completeness as well as creativity of the written answer.

**1.** How could you measure glucose on-line in a bioreactor that contains many other components including high concentrations of lactic acid and glutamic acid? The measurement must be obtained every 5 minutes. The bioreactor is producing the protein, interferon-beta-1-alpha, a family of proteins used to treat multiple sclerosis. The reactor contains cells along with all the byproducts (waste) from cellular respiration. Be sure to discuss your rationale for the method (or methods) you have chosen. A complete description of the analytical chemistry (instrumentation, how it works, interference considerations, etc.) is required for full credit.

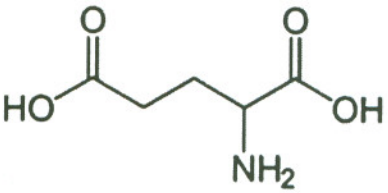
Structures:




Glucose

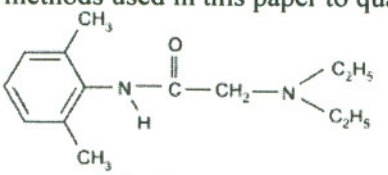



Lactic Acid

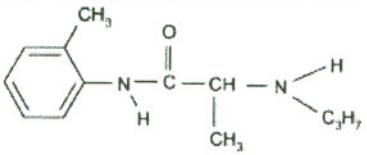



Glutamic acid

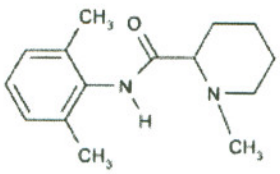
**2.** A recent paper has been published: Abdel-Rehim, Mohamed. "New trend in sample preparation: on-line microextraction in packed syringe for liquid and gas chromatography applications I. Determination of local anaesthetics in human plasma samples using gas chromatography-mass spectrometry." Journal of Chromatography, B: Analytical Technologies in the Biomedical and Life Sciences (2004), 801(2), 317-321. In this paper, the author discusses a new device for performing solid-phase extraction for the anesthetics shown below. Answer the following questions regarding the figures of merit and calibration methods used in this paper to quantify these analytes from human plasma samples.



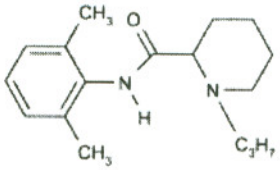

Lidocaine




Prilocaine




Mepivacaine




Ropivacaine

The following information is cut directly from the methods section. "Stock solutions of the analytes in methanol were prepared. Spiked plasma samples were prepared by adding a few microlitres of analyte standard to 1.0 ml plasma, after which 50 µl of the internal standard, [$^2H_7$] ropivacaine. (The [$^2H_7$] means there are 7 hydrogens that have been substituted with deuterium) (25 µM) was added. The concentration range of standard curves was between 5 and 2000 nM."

a. What would be the purpose for spiking plasma with analyte standards?
b. Why is an internal standard used in this method? Why does it contain deuterium?
c. Below are two chromatograms run in SIM mode using a mass spectrometer for spiked samples and blanks. Discuss why the authors would prepare blanks and what purpose they serve for this analysis. Why is SIM mode used? What advantage does it serve?

**Figure 3**

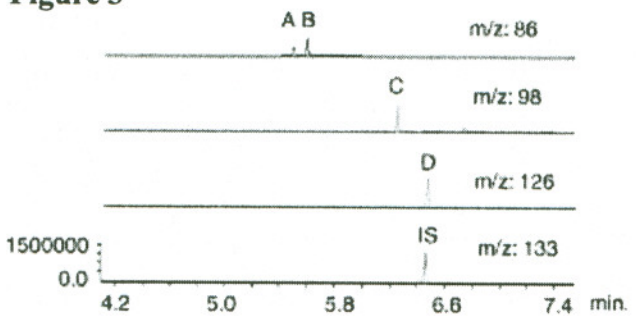



**Figure 4**

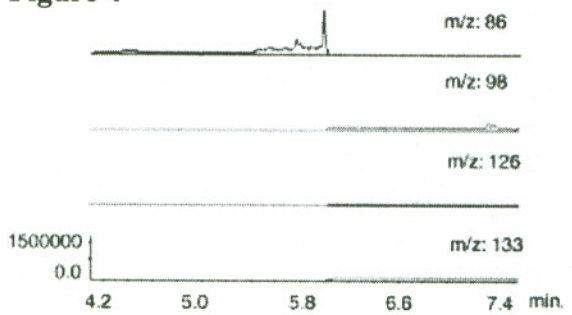
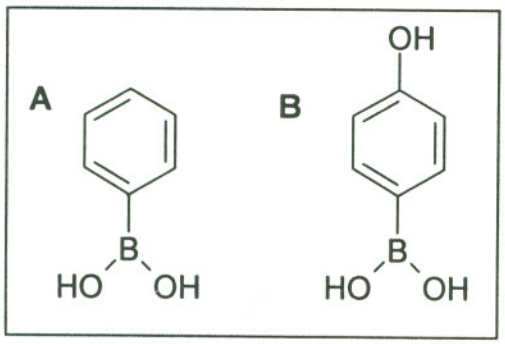



Fig. 3. Mass chromatograms obtained from human plasma spiked with analytes (*m/z*: 86, 86, 98, and 126 for prilocaine, lidocaine, mepivacaine and ropivacaine, respectively, 800 nM each) and I.S. (*m/z*: 133, 1250 nM).

Fig. 4. Mass chromatograms obtained from human blank plasma.

**3.** For the two compounds that contain boron (which are used in cancer treatment), describe the <u>most selective</u> method for quantifying these analytes in a complex matrix such as blood. Be sure to discuss your rationale for the method (or methods) you have chosen. A complete description of the analytical chemistry (instrumentation, how it works, interference considerations, etc.) is required for full credit.



**4.** A 0.1 M sodium citrate buffer, has been prepared to pH 5.0. Citric acid is a triprotic acid. For citric acid, the $pK_a$ values are: 3.128, 4.761, and 6.396.

a) Denote the two predominant (e.g., highest concentration) forms of citric acid at this pH (Use $H_3A$, $H_2A^-$, etc to denote the forms with highest concentration)

b) In this solution, what are the concentrations of these two predominant forms?

c) If 100 mL of this 0.1 M citric acid buffer, pH 5, is added to 100 mL of 0.025 M $HNO_3$, what is the pH of this new solution?